

Name :

Roll No. :

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Tutorial 3

VIth Semester

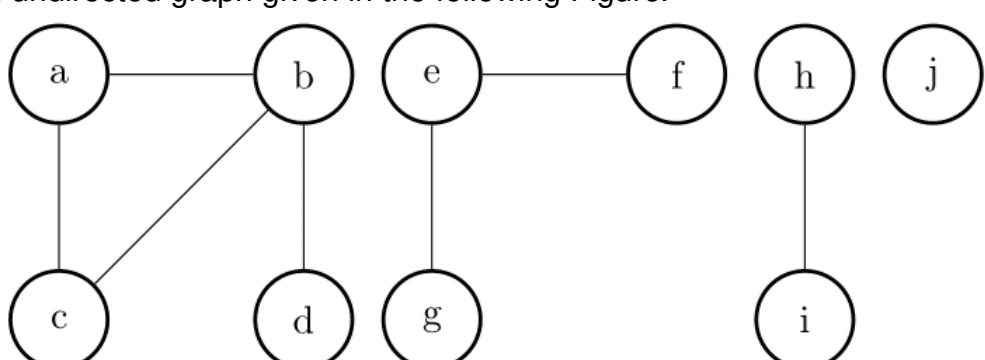
CST306 - ALGORITHM ANALYSIS AND DESIGN

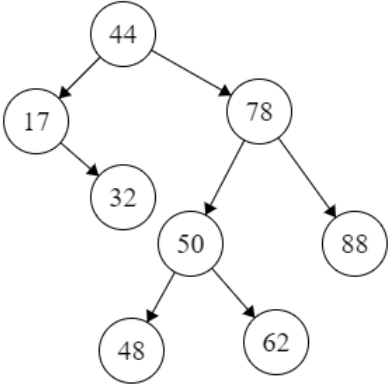
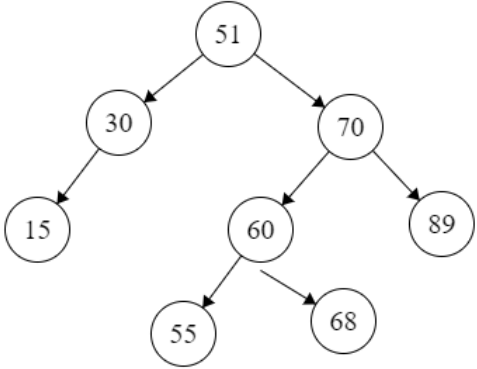
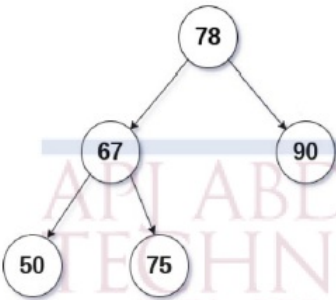
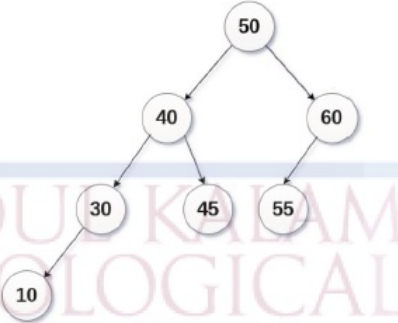
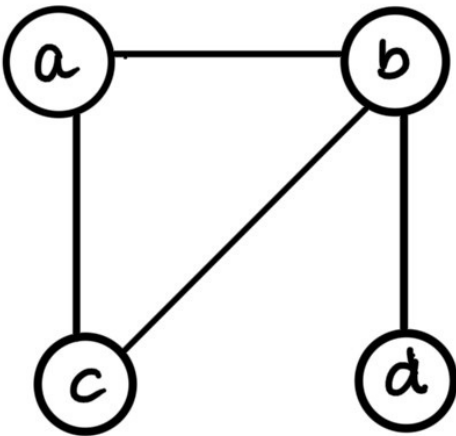
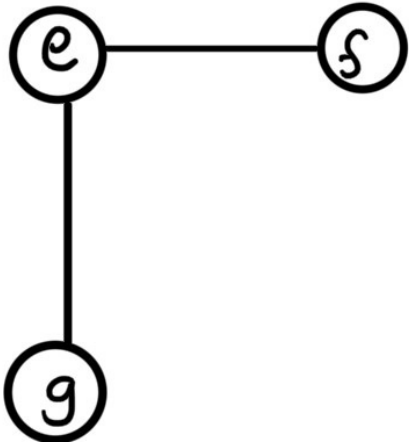
AI & DS 2023

Total Mark: 15

Total Time: 2 Hrs

Course Outcome (CO)	
After completion of this course, the students will be able to	
CO3	Illustrate Graph traversal algorithms, it's applications and Advanced Data structures like AVL trees and Disjoint set operations.

No.	Question	Marks	CO	BL
1.	Define AVL tree. Construct an AVL tree by inserting the keys: 44, 17, 32, 78, 50, 88, 48, 62, 54 into an initially empty tree. Write clearly the type of rotation performed at the time of each insertion.	2 1/2	CO3	3
2.	What are the operations supported by Disjoint Data Structure? Explain the working of Disjoint Set Data Structure for computing Connected Components of an undirected graph given in the following Figure. 	2 1/2	CO3	3

3.	Perform the following operations in the given AVL trees. 1) Insert 54 in Tree 1 2) Delete 15 from Tree 2.  	2 1/2	CO3	3
4.	Construct AVL tree by inserting following elements appeared in the order. 21, 26, 30, 9, 4, 14, 28, 18,15	2 1/2	CO3	3
5.	Perform the following operations in the given AVL trees. i) Insert 70 ii) Delete 55  	2 1/2	CO3	3
6.	Give an algorithm for computing connected components in a graph using Disjoint Set data structure operations. Explain the working of this algorithm on the graph given below.  	2 1/2	CO3	3

Prepared by	Verified by	Approved by

LEVEL - Bloom's Taxonomy Level	
Level 3	Applying (P)